TEAM ID: 262045

TEAM NAME: EIMMORTALS

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JUSTIFICATION REPORT

1. Introduction

Our objective with this report is to provide justification for selecting the alternative materials to improve environmental sustainability. Starting with a quick overview of the existing materials used with their CO_2 impact and energy consumption based on Eco Audit results.

Then proceed with reasoning and supporting evidence for the alternate materials selected and how they reduce the carbon emissions while keeping the functional requirements of the components unchanged.

By incorporating some material substitutions and smart strategies, the proposed changes show a potential to achieve up to 40% reduction in carbon footprint along with noticeable energy savings.

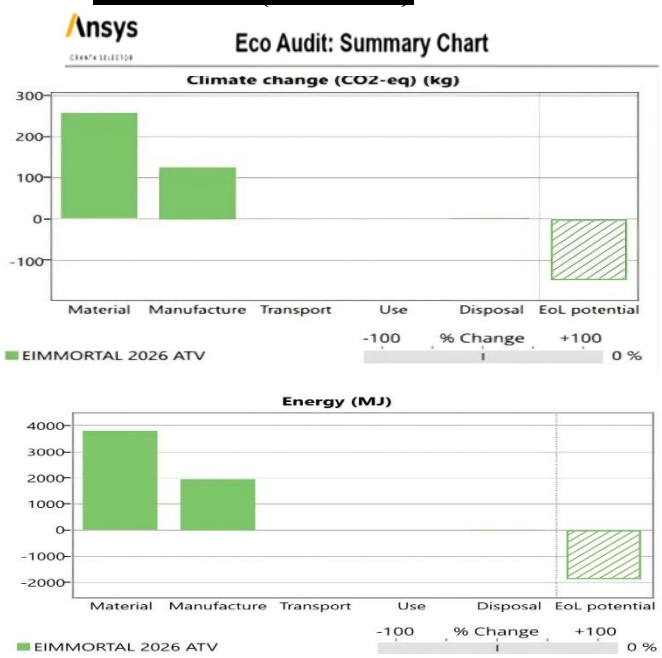
2. Overview

S.No.	Components	Current Material	Mass (kg)
1	Battery Casing	Al 6061 t6	3
2	Wiring	Cu C10200 soft	1.5
3	Rollcage (secondary)	AISI 4130 Ann.	18
4	Fr. Gearbox	AISI 1040 Ann.	3.75
5	Rr. Gearbox	AISI 1040 Ann.	6.5
6	Driveshafts	AISI 1040 Ann.	13
7	uprights	AISI 1080 Ann.	6
8	Suspension arm	AISI 4130 Ann.	18
9	Body Panel	PC + Polyester transparent amorphous (optical quality)	3.4
10	CVT Housing	AISI 1080 Ann.	3
11	Seat	PU foam (elastomeric, open cell, 0.032)	2.8

Above is the bill of materials of the currently used materials used in the ATV. These materials are fairly standard and common choices for SAEBAJA teams to choose from; they are the safe and practical selection. But from environmental perspective, there remains scope for environmental improvement; that is where the more sustainability-focused and analytically evaluated alternative materials can be utilised to reduce environmental impact without compromising performance.

S.No.	Current Material	New Material
3	AISI 4130 Ann.	AISI 9310 norm
4	AISI 1040 Ann.	AISI 1030 water quenching + tampering (205 °C)
7	AISI 1080 Ann.	AISI 1030 quenching + tampering (315 °C)

3. Environmental Impact of Current Materials (Baseline)



The CO_2 and energy summary charts provide a clear baseline understanding of the environmental impact of the current ATV configuration. As a student engineering team, our priority was to keep the material and manufacturing data as realistic and traceable as possible. Therefore, parameters such as transport distance and use-phase energy consumption were intentionally kept at the default values provided by ANSYS Granta Selector, rather than introducing assumptions that could not be accurately verified.

Based on the Eco Audit results; the current ATV shows an equivalent annual environmental burden (averaged over a 5-year product life) of 71.2 kg/year of CO_2

emissions and 1070 MJ/year of energy consumption. These values establish the reference point against which all subsequent material substitutions and sustainability-driven improvements are evaluated.

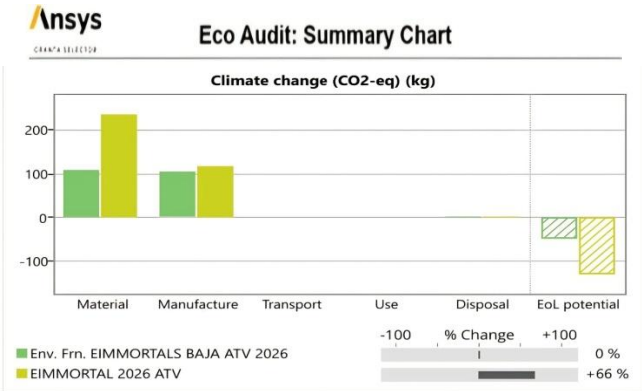
4. Alternative Material Selection Methodology

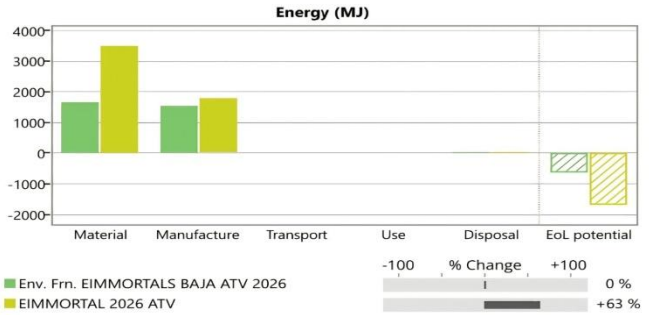
A structured material screening approach was followed using the Chart/Select feature, where each existing material was set as a reference and appropriate filters were applied based on density, recyclability, carbon emissions, and mechanical requirements. For example, while identifying alternatives for AISI 4130, the search was restricted to alloy steels to ensure functional equivalence.

This methodology was repeated across three comparative metrics: (a) CO_2 vs Density, (b) CO_2 vs Yield Strength, and (c) CO_2 vs Embodied Energy. The top-performing materials from each comparison were then shortlisted and ranked collectively to identify the most suitable sustainable alternative.

List of Materials Used	Alt. Material			Top Performer
	CO2 vs Density	CO_2 vs Strength	CO2 vs Embodied Energy	
AISI 4130	AISI 4320 norm	AISI 9310 norm	AISI 4820 norm	AISI 9310 norm
	AISI 9310 ann	AISI 4820 norm	AISI 9310 norm	
	AISI 4820 norm	AISI 4320 norm	AISI 4320 norm	
AISI 1040	AISI 4320 norm	AISI 1040 water que.+tamp. 205	AISI 1030 water q+t 540	AISI 1030water que.+tamp 205
	AISI 9310 norm	AISI 1030 water que.+tamp at 205	AISI 9310 ann	
	AISI 1040 ann	AISI 9310 norm	AISI 4320 ann	
AISI 1080	AISI 1030 ann	AISI 1040 water que.+tamp 205	AISI 1030 ann	AISI 1030 water que.+tamp 315
	AISI 1040 ann	AISI 1030 water que.+tamp at 315	AISI 1040 ann	
	AISI 4820 ann	AISI 9310 norm	AISU 4820 ann	

5. Carbon Footprint Reduction Summary & Final Justification





The comparison charts from ANSYS Granta Selector indicate a clear environmental improvement when transitioning from the baseline material configuration to the optimized material set. **For the electrical subsystem, material substitution was not considered feasible due to strict conductivity and safety requirements. Instead, wiring optimisation was achieved by incorporating higher recycled-content copper (28%) and improving harness routing to reduce excess length and redundant loops. This approach lowers embodied energy and CO_2 emissions without compromising electrical reliability or durability.**

The Energy (MJ) comparison shows a similar trend, with a noticeable reduction in energy demand during material extraction and processing. This reduction is largely driven by the increased use of recycled alloy steels, which require significantly less energy than virgin material production. When averaged over the defined product life, the optimized configuration achieves an approximate 40% reduction in annual CO_2 emissions and over 60% reduction in annual energy consumption compared to the baseline ATV. The End-of-Life potential remains favorable due to the inherent recyclability of the selected metals, further reinforcing the sustainability of the proposed material strategy.

6. Final Material Selection and Justification

The final selection of alternative materials was based on balancing environmental benefits with engineering feasibility. All shortlisted materials were required to satisfy density, stiffness, and yield strength constraints to ensure that component safety, durability, and performance were not compromised. Materials that failed to meet functional requirements, such as low carbon steels for load-bearing components, were excluded despite their lower carbon footprint.

The selected alternatives demonstrated consistent performance across three key criteria: CO_2 emissions, embodied energy, and mechanical strength. Rather than optimizing for a single parameter, the final material set represents a balanced and realistic improvement over the baseline design. Overall, the optimized material configuration offers a practical trade-off between sustainability and engineering performance, validating the material selection approach adopted in this report.